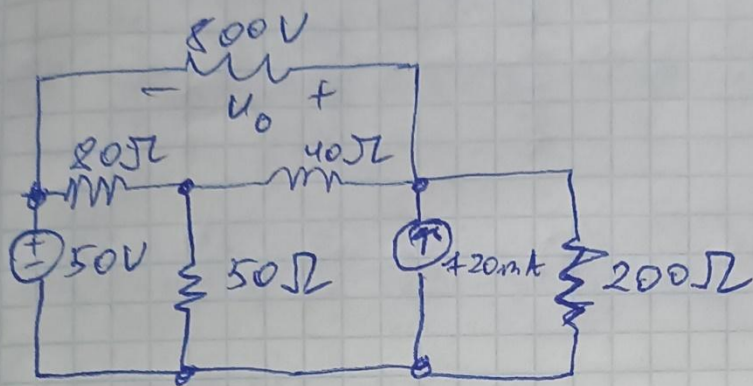


[P 132 N4,23]

#5



$$U_0 = V_2 - 50$$

$$\begin{cases} \frac{V_1 - 50}{80} + \frac{V_1}{50} + \frac{V_1 - V_2}{40} = 0 \\ \frac{20}{40} \frac{V_2 - V_1}{40} - \cancel{0.42} + \frac{V_2}{200} + \frac{V_2 - 50}{80} = 0 \end{cases}$$

$$\frac{3V_1 - 250}{40} + \frac{8V_1}{40} + \frac{10V_1 - 10V_2}{40} = 0$$

$$23V_1 - 10V_2 = 250$$

$$\frac{20V_2 - 20V_1}{80} - \frac{45.8}{80} + \frac{4V_2}{80} + \frac{V_2 - 50}{80} = 0$$

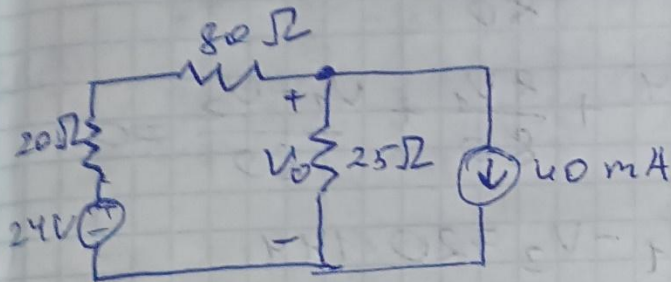
$$-20V_1 + 25V_2 = 75.8 + 50$$

$$\begin{cases} -20V_1 + 25V_2 = 125.8 \\ 23V_1 - 10V_2 = 25 \end{cases}$$

$$\begin{aligned} V_2 &= 53.2V \\ V_1 &= 34V \end{aligned}$$

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#1



25Ω:

$$\frac{V_0 - 24}{20 + 80} + \frac{V_0}{25} + 0.04 = 0$$

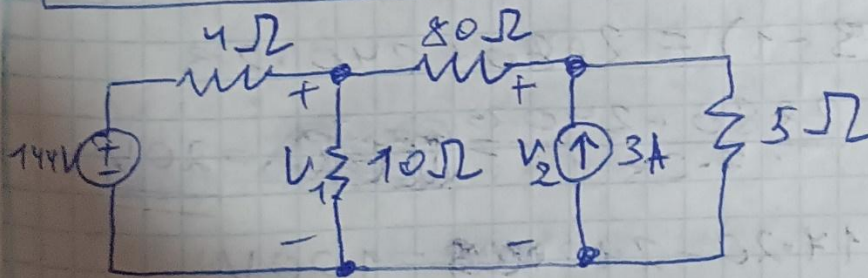
$$\frac{V_0 - 24}{100} + \frac{4V_0}{100} = -4 \cdot 10^{-2}$$

$$5V_0 - 24 = -4$$

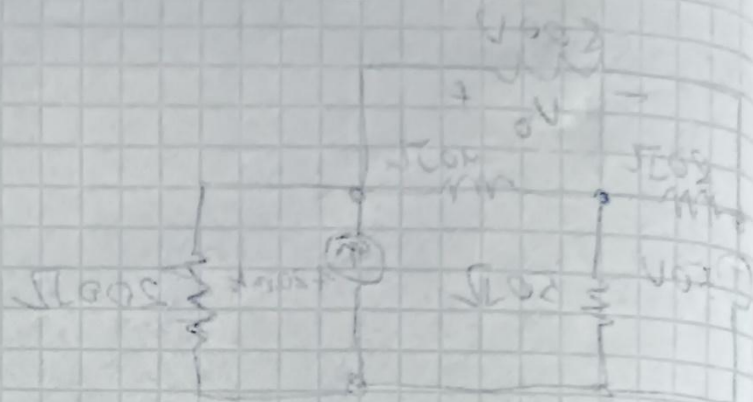
$$5V_0 = 20$$

$$\underline{V_0 = 4 \text{ V}}$$

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#6 $V_0 = V_2 - 50 = 3,2V$



$$0 = \frac{V_0 - 10}{500} + \frac{V_0}{500} + \frac{V_0 - 10}{500}$$

$$0 = \frac{V_0 - 10}{500} + \frac{V_0}{500} + \frac{V_0 - 10}{500}$$

$$0 = \frac{V_0 - 10}{500} + \frac{V_0}{500} + \frac{V_0 - 10}{500}$$

$$0 = \frac{V_0 - 10}{500} + \frac{V_0}{500} + \frac{V_0 - 10}{500}$$

$$0 = \frac{V_0 - 10}{500} + \frac{V_0}{500} + \frac{V_0 - 10}{500}$$

$$0 = \frac{V_0 - 10}{500} + \frac{V_0}{500} + \frac{V_0 - 10}{500}$$

$$V_0 = 3,2V$$

$$\#4 \quad 6 \cdot 120 - 5V_2 = 240$$

$$5V_2 = 720 - 240 \Rightarrow V_2 = \frac{480}{5} \Rightarrow \underline{V_2 = 96V}$$

checking:

$$P_{40\Omega} = \frac{(120)^2}{40} = 360W$$

$$P_{8\Omega} = \frac{(120 - 96)^2}{8} = 72W$$

$$P_{80\Omega} = \frac{(96)^2}{80} = 115.2W$$

$$P_{120\Omega} = \frac{(96)^2}{120} = 76.8W$$

$$P_{6A} = -(6)(120) = -720W$$

$$P_{1A} = 1 \cdot 96 = 96W$$

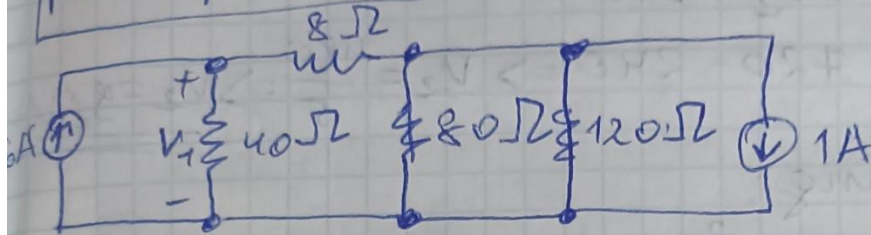
$$\sum P_{ABS} = \cancel{360} + 72 + 115.2 + 76.8 + 96 = 720W$$

$$\sum P_{dev} = 720W \quad \checkmark$$

$$\underline{V_1 = 120V, V_2 = 96V}$$

p. 131 / 4.13

#3



$$\begin{cases} -6 + \frac{V_1}{40} + \frac{V_1 - V_2}{8} = 0 \\ \frac{V_2 - V_1}{8} + \frac{V_2}{80} + \frac{V_2}{120} + 1 = 0 \end{cases}$$

$$\frac{V_1}{40} + \frac{5V_1 - 5V_2}{40} = 6$$

$$6V_1 - 5V_2 = 240$$

$$V_2 - V_1 + \frac{V_2}{10} + \frac{V_2}{15} = -8$$

$$5V_2 - 5V_1 + \frac{V_2}{2} + \frac{V_2}{3} = -40$$

$$30V_2 - 30V_1 + 3V_2 + 2V_2 = -240$$

$$35V_2 - 30V_1 = -240$$

$$\begin{cases} -5V_2 + 6V_1 = 240 \\ -35V_2 + 42V_1 = 240 \end{cases}$$

$$-35V_2 + 42V_1 = 240$$

$$12V_1 = 240 \Rightarrow V_1 = \underline{\underline{20V}}$$

$$\#2 \quad \frac{V_1 - 144}{4} + \frac{V_1}{10} + \frac{V_1 - V_2}{80} = 0$$

$$\frac{20V_1 - 20 \cdot 144}{80} + \frac{8V_1}{80} + \frac{V_1 - V_2}{80} = 0$$

$$20V_1 + 8V_1 + V_1 - V_2 = 20 \cdot 144$$

$$29V_1 - V_2 = 2880$$

$$-3 + \frac{V_2 - V_1}{80} + \frac{V_2}{5} = 0$$

$$\underline{V_2 - V_1 + 16V_2 = 3 \cdot 80}$$

$$\begin{cases} 17V_2 - V_1 = 240 \end{cases} \cdot 29$$

$$\begin{cases} 29V_1 - V_2 = 2880 \end{cases}$$

$$+ \quad 17V_2 \cdot 29 - 29V_1 = 240 \cdot 29$$

$$- V_2 + 29V_1 = 2880$$

$$V_2(17 \cdot 29 - 1) = 2880 + 240 \cdot 29$$

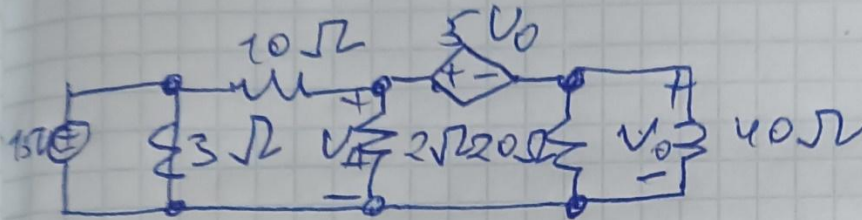
$$V_2 = \frac{2880 + 240 \cdot 29}{17 \cdot 29 - 1} = \underline{\underline{20V}}$$

$$V_1 = 17 \cdot 20 - 240 = \underline{\underline{100V}}$$

$$\underline{\underline{V_1 = 100V, V_2 = 20V}}$$

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#4



$$\frac{V_A - 15}{10} + \frac{V_A}{20} = \frac{V_A - 5V_A}{20} + \frac{V_A - 5V_A}{40} = 0$$

$$V_A = 5V$$

$$V_0 = V_A - 5V_A = -20V$$

$$V_0 = -20V$$